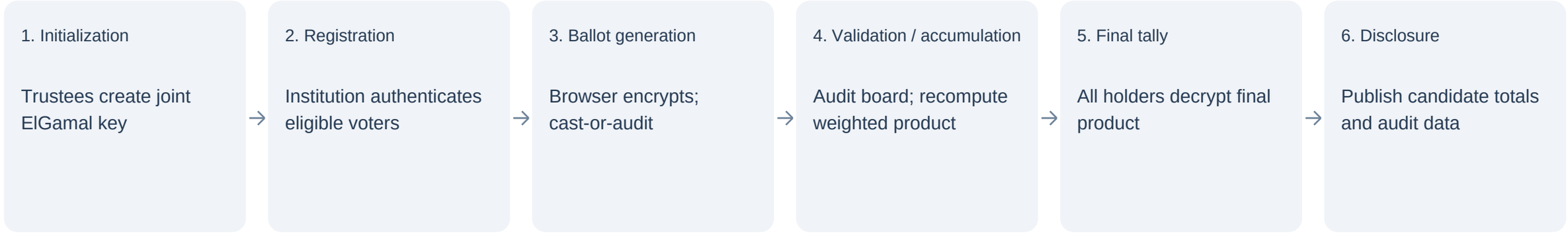
Helios 2.0 UCL | Architecture and trust boundary

2009 UCL deployment, not Helios 1.0 mixnet

Selected version only. Public recomputation is a valid consistency check; no state SNARK is required.



Execution: Web/server and offline audit; no blockchain mechanism established.

Key custody: Distributed key holders.

Tally computation: Key holders and tally service.

Full tally visibility: Public.

Public verification: Ballot and decryption audits.

Trust assumptions: Authentication, voter auditing and secret key custody; compromised clients remain a limitation.

Evidence: Section 2.2, PDF p.3; Section 2.3, PDF p.4; Sections 2.1,2.6, PDF pp.3,5; Sections 3.4-3.6, PDF pp.10-11; Section 3.5, PDF pp.10-11; Table 1, PDF p.12

https://usenix.org/events/evtwote09/tech/full_papers/adida-helios.pdf

Belenios | Architecture and trust boundary

Specification 3.0, homomorphic mode with Pedersen trustees

Selected version only. Public recomputation is a valid consistency check; no state SNARK is required.

Execution: Voting server and public hash-chain archive.

Key custody: Configured trustees.

Tally computation: Trustees and voting server.

Full tally visibility: Public in selected mode.

Public verification: Recompute retained-set tally; verify proofs.

Trust assumptions: Credential/trustee role configuration and monitored board; no unconditional server-honesty claim.

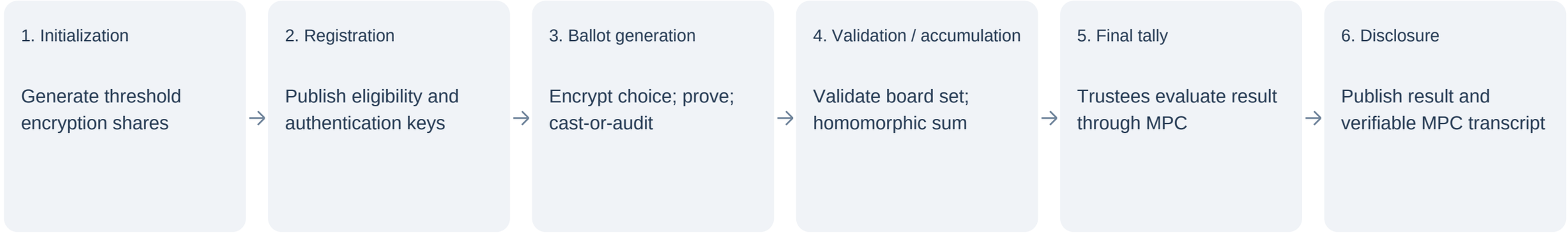
Evidence: Section 3.1.1, PDF pp.4-5; Section 3.1, PDF pp.3-4; Section 3.2, PDF p.5; Section 3.5, PDF pp.6-7; Section 3.4, PDF p.6; Sections 3.4-3.5, PDF pp.6-7

https://www.belenios.org/specification.pdf

Ordinos | Architecture and trust boundary

EuroS&P 2020 extended author version

Selected version only. Public recomputation is a valid consistency check; no state SNARK is required.



Execution: Authenticated bulletin board and trustee MPC.

Key custody: Threshold trustees; prototype central setup.

Tally computation: Trustees jointly evaluate result.

Full tally visibility: Hidden under P1-P4, including from subthreshold trustees.

Public verification: Judging procedure over public transcript.

Trust assumptions: P1-P4, t-1 corrupt trustees, protected devices, honest board/scheduler and risk-avoiding adversary; prototype trusted setup distinguished.

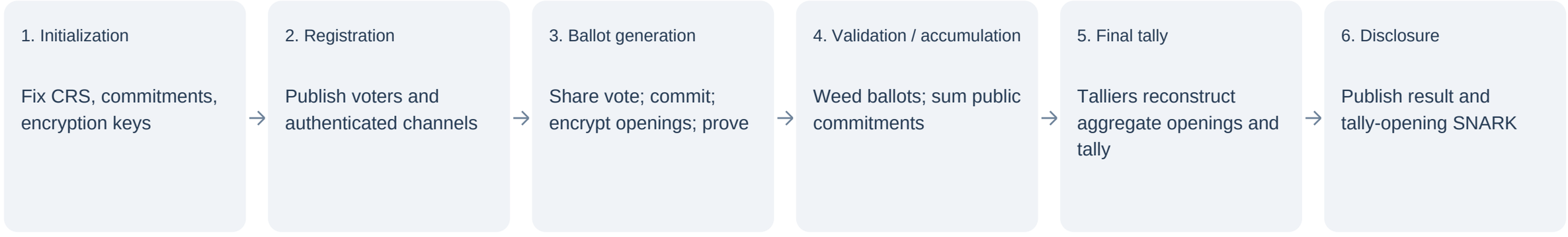
Evidence: Section 2, PDF pp.6-7; Section 2, PDF p.7; Section 2, PDF pp.8-9; Sections 2,7, PDF pp.8-9,20; Sections 2,4, PDF pp.8-10

https://publ.sec.uni-stuttgart.de/kuestersliedtkemuellerrauschvogt-eurosp-2020.pdf

Kryvos | Architecture and trust boundary

CCS 2022, 15-page author version

Selected version only. Public recomputation is a valid consistency check; no state SNARK is required.



Execution: Public board and private tallier exchange.

Key custody: Each tallier has an encryption key.

Tally computation: All talliers; designated proof producer.

Full tally visibility: Talliers learn full tally; public learns result.

Public verification: Commitment sums and result SNARK.

Trust assumptions: Public privacy: all talliers honest; internal privacy: at most nt-1 corrupt; honest BB/scheduler; honestly generated CRS.

Evidence: Section 3.1, PDF p.6; Section 3.1, PDF pp.6-7; Section 3.1, PDF p.7

https://publ.sec.uni-stuttgart.de/huberkuesterskripsliedtkemuellerrauschreisert-ccs-2022.pdf

Emami et al. | Architecture and trust boundary

JISA 79 (2023) 103645; full text pending

PENDING ORIGINAL FULL TEXT - no inferred security properties or implemented baseline

Execution: Publisher reports off-chain computation/on-chain coordination; detailed mapping pending.

Key custody: Unknown.

Tally computation: Unknown.

Full tally visibility: Unknown.

Public verification: Unknown pending full protocol.

Trust assumptions: Pending original full text; weak receipt-freeness is not coercion resistance.

Evidence: Unavailable

https://www.sciencedirect.com/science/article/pii/S2214212623002296

VeriVoting-Batch prototype | Architecture and trust boundary

5f170c5, G10-C native plus local EVM fallback

NATIVE + LOCAL EVM FALLBACK - exact SnarkPack EVM remains incomplete

1. Initialization	2. Registration	3. Ballot generation	4. Validation / accumulation	5. Final tally	6. Disclosure
Rebuild certified research DKG context	Bind registration close and weights	Weighted encrypt; verify ballot proofs	Native SP+state; EVM individual+state	Freeze CT2; private CP; new P3	Storage-derived P3; write Winner once

Execution: Native plus local Prague EVM individual-proof fallback.

Key custody: Central research bootstrap; five CP worker processes.

Tally computation: Authorized combiner.

Full tally visibility: Combiner learns tally; public output excludes it.

Public verification: Authenticated frozen-state P3 and once-only storage.

Trust assumptions: Confidentiality-trusted combiner, external identity roots; deterministic research setup/Fake SRS; T4/T5 not proved.

Evidence: g10c_complete_native.rs; common/g6_materials.rs; common/g6_batches.rs; common/g2e_receiver.rs; g10c_winner_chain.mjs; scripts/g10c_chain_audit.mjs

https://github.com/fate-st/verivoting/tree/5f170c5a904d825a99c96fd45558a743f2c4e1ea